

If I Change Your Eyes, Are You Still You? Identity Thresholds in Deepfake Perception

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Abstract

Deepfake harm arises not from technical manipulation alone, but from the moment viewers perceive the synthesized face as a specific victim. Yet the threshold at which this identity perception occurs remains undefined, creating gaps in both detection systems and legal frameworks. We investigate the conditions under which synthesized faces are perceived as target individuals and examine alignment between human perception and computational similarity models. Through a study with 102 participants evaluating 162 manipulated face stimuli, we identify three key findings: (1) eyes serve as the dominant cue for identity perception; (2) a distinct identity threshold emerges when eyes are combined with one additional feature; and (3) human-AI misalignment peaks at this threshold, where humans perceive the victim but AI models do not. These findings highlight the need for perception-aligned detection systems and victim-centered regulatory approaches in addressing deepfake harms.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Empirical studies in HCI**; • **Computing methodologies** → *Artificial intelligence*.

Keywords

Deepfake, Identity perception, Human-AI alignment, Face recognition, Identity threshold

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1 Introduction

In January 2024, deepfake sexually explicit images of pop star Taylor Swift were widely circulated on X [9]. Such incidents illustrate how deepfakes can cause harm not only by resembling a person, but also by leading viewers to believe that the depicted individual is the victim. While attacks on public figures are devastating, they are, in principle, detectable: content moderation systems can rely on face recognition to flag and remove unauthorized content featuring known celebrities. However, the rapid democratization of deepfake technology has fundamentally shifted the threat landscape. Today, deepfake pornography accounts for 98% of all deepfake content [6], and targets are increasingly ordinary individuals, including middle and high school students [24]. Unlike celebrities, these victims lack reference databases, rendering face recognition-based content moderation ineffective for a growing majority of those harmed by deepfakes.

In response to this crisis, several jurisdictions have enacted legislative measures targeting non-consensual deepfakes. In the United States, the “TAKE IT DOWN Act” was signed into law in May 2025, criminalizing the intentional distribution of non-consensual intimate images, including AI-generated deepfakes, at the federal level [25]. Comparable provisions have also been adopted in U.S. states (e.g., California and Virginia) and in the United Kingdom (via amendments to the Online Safety Act) [10, 11]. Across these efforts lies a shared premise: harm and liability hinge on whether content is perceived as depicting a real, identifiable individual.

Current legal frameworks struggle to operationalize this discrepancy. Although statutes prohibit the unauthorized use of an individual’s likeness, identity in deepfakes is not a binary property determined by whether an entire face is used. Instead, identity perception lies on a continuous spectrum of resemblance: viewers may recognize salient cues (e.g., the eyes) and attribute identity to a victim even without a full facial match. In cases of deepfake sexual exploitation, harm arises not from technical manipulation alone,

but from the moment manipulated content is perceived as depicting a specific individual. This recognition-dependent nature of harm poses a shared challenge for both regulation and detection: *at what point does a synthesized face cross the threshold to be perceived as the victim?*

Prior work has addressed facets of this question across three streams. Cognitive psychology has established that humans rely differentially on specific facial components, with the eye region playing the most critical role [13, 18]; face processing also differs by familiarity, with internal features dominating for familiar faces and external features for unfamiliar faces [5, 27]. Computer vision has quantified facial resemblance using embedding-based models such as ArcFace [3], now the de facto standard for evaluating identity preservation in deepfake generation, yet detection research remains optimized for binary authenticity classification [17, 26]. Human-AI alignment research has largely focused on language models or decision-making trust [21], with face recognition studies confined to security contexts [4, 14].

Despite these advances, existing work remains siloed and does not ask when a synthesized face is perceived as a specific individual—the **identity threshold**. As a result, content that computational models classify as a different identity may still be perceived by humans as the victim, limiting the effectiveness of detection. We therefore take an intersectional approach spanning cognitive mechanisms of identity recognition, deepfake similarity measurement, and human-AI perception alignment.

To bridge this gap between technical classification and human perception, we empirically investigate where deepfake identity boundaries emerge. Rather than treating identity as a binary property, we investigate the point at which increasing facial manipulation leads human observers to attribute identity to a specific individual, and how this perceptual threshold aligns (or misaligns) with computational similarity metrics. Specifically, we pose two research questions: (**RQ1**) At what level of facial manipulation do humans begin to perceive a synthesized face as the target individual? (**RQ2**) To what extent do computational similarity metrics diverge from human identity perception across these conditions?

This study presents an exploratory first step toward addressing these foundational questions by empirically characterizing how identity is perceived in deepfake images. We identify three key findings. First, the eye region serves as the dominant cue for human identity perception. Second, a distinct identity threshold emerges when the eye region is combined with just one other feature, at which point perception crosses the ambiguity boundary and begins to transition toward the target identity. Third, systematic misalignments exist between human perception and computational similarity metrics, particularly around this threshold. We conclude by outlining how perception-aligned detection should prioritize eye-region changes and feature interactions, and how the two-feature identity threshold can inform victim-centered harm criteria in emerging deepfake regulation.

2 Exploratory Study Design

2.1 Stimuli Design

Facial Feature Selection. We selected facial features for manipulation based on prior face recognition and deepfake literature,

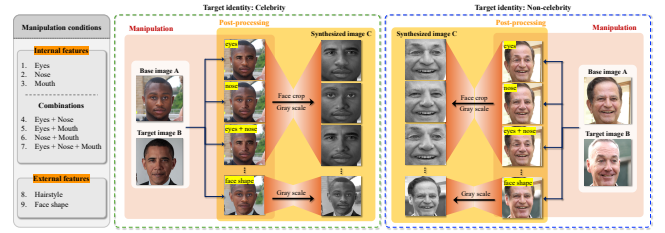


Figure 1: Illustration of the stimuli generation pipeline and the nine manipulation conditions.

grouping them into internal and external cues. Internal features included eyes, nose, and mouth, which form the canonical T-shaped facial configuration [23]. Because the eye region is highly diagnostic for identity [13] and eyebrows provide comparable identity information in close anatomical proximity [19], we manipulated eyes and eyebrows as a single unit. We included the mouth given its frequent manipulation in lip-sync and face reenactment deepfakes [15]. External features comprised hairstyle and face shape, which can substantially alter perceived identity, particularly for unfamiliar faces [5, 22].

Based on these considerations, we established nine manipulation conditions: three single internal features (eyes, nose, mouth), four internal feature combinations (eyes+nose, eyes+mouth, nose+mouth, eyes+nose+mouth), and two external features (hairstyle, face shape). The focus on internal feature combinations reflects the fact that most deepfake generation techniques primarily rely on substituting internal facial features.

Stimuli Construction. We selected original face images from the Flickr-Faces-HQ (FFHQ) dataset [7], which permits redistribution and modification. We define base image A as the recipient face and target image B as the source identity whose features are composited onto A; in deepfake victimization scenarios, B corresponds to the victim. Targets B included nine celebrities and nine non-celebrities to capture familiarity-related processing differences [5]. Each A–B pair was matched on gender, age range, and ethnicity to ensure natural synthesis and reduce confounds in identity perception. We manually created synthesized image C in GIMP by compositing features from B onto A. To minimize perceptual variation, all images were converted to grayscale, and internal-feature conditions were cropped to the facial region to exclude external cues. The final stimulus set comprised 162 images (9 manipulation conditions $\times$ 18 A–B pairs). Figure 1 illustrates the stimuli generation pipeline and example outputs.

2.2 Procedure

Human Perception Measurement. Human identity perception was measured via an online survey. Sample size was determined based on power analysis for a within-subject design (medium effect size, $d = 0.5$) and sample sizes from comparable face perception studies [$N=80$; $N=82$] [2, 8], resulting in a final sample of 102 participants. The survey was conducted from December 26, 2025 to January 10, 2026, with participants receiving approximately \$7 compensation upon completion. In each trial, participants viewed base image A, target image B, and synthesized image C simultaneously, then indicated who they perceived C to be via forced-choice selection of either A or B. Each participant evaluated all nine manipulation

Table 1: Effect of facial feature manipulation on identity perception. B% = observed target selection rate across all participants. β and p-values estimated from binomial GLMM with participant and stimulus as random effects. Combination conditions are not modeled separately due to collinearity with individual features.

Manipulation	Eyes	Nose	Mouth	Face shape	Hairstyle	Eyes+Nose	Eyes+Mouth	Nose+Mouth	Eyes+Nose+Mouth
B% (Overall)	39.9%	2.7%	2.8%	1.1%	4.1%	70.5%	64.9%	18.6%	87.7%
B% (Celeb)	36.2%	4.2%	3.4%	1.6%	3.3%	69.8%	59.5%	21.0%	88.4%
B% (n/Celeb)	43.5%	1.1%	2.1%	0.6%	5.0%	71.2%	70.3%	16.2%	87.0%
β (Overall)	4.78	1.41	1.51	-0.14	1.87	–	–	–	–
β (Celeb)	4.21	1.84	1.45	0.41	1.38	–	–	–	–
β (n/Celeb)	4.78	1.41	1.51	-0.14	1.87	–	–	–	–
p	<.001	<.001	<.001	.537	<.001	–	–	–	–

conditions for 18 A-B pairs, with an average completion time of 15–20 minutes.

AI Similarity Measurement. To analyze alignment between human perception and AI judgment, we measured facial similarity using models widely adopted in deepfake research. We employed ArcFace [3], FaceNet [20], VGG-Face [12], and OpenFace [1], all widely adopted for measuring identity similarity in deepfake research. We also used CLIP [16], a vision-language model that has recently gained attention for its generalization performance in deepfake detection despite not being face-specialized. For each model, we computed the similarities between the synthesized image C and the base image A (sim_A) and between C and the target image B (sim_B). When $sim_B > sim_A$, the model was deemed to have identified C as B.

2.3 Analysis

Human Perception Analysis. For each stimulus, we calculated the rate at which participants selected B (B_rate). Our focus on perception as target B (rather than A) reflects the nature of deepfake harm—damage occurs when a synthesized face is recognized as the target individual. This directly addresses RQ1: under what manipulation conditions is a synthesized face perceived as the target individual? Applying the majority rule, stimuli with $B_rate > 0.5$ were classified as perceived as target B.

Human-AI Alignment Analysis. We compared human perception data with AI model judgments, conceptually distinguishing four outcome types: (1) Agreement-Positive (both human and AI select B); (2) Agreement-Negative (both human and AI select A); (3) Human-only (human selects B, AI selects A); (4) AI-only (AI selects B, human selects A). Category (3) identifies conditions where AI-based detection may fail to capture actual harm; category (4) indicates conditions where AI over-detection may undermine regulatory legitimacy. To address RQ2—under what conditions does human-AI misalignment occur—we primarily report condition-wise target selection rates (Fig. 2b) as a compact summary of alignment and mismatch patterns.

3 Results

3.1 Eyes as a Dominant Cue for Identity Perception

To address RQ1, we applied a binomial Generalized Linear Mixed Model (GLMM) with manipulation condition as a fixed effect and participant and stimulus as random effects. Eye manipulation exhibited the strongest effect on target identity perception ($\beta = 4.78$, $p < .001$), followed by hairstyle ($\beta = 1.87$), mouth ($\beta = 1.51$), and nose

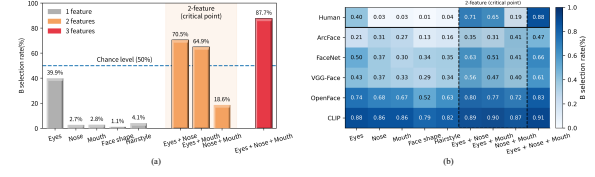


Figure 2: Identity threshold and human-AI misalignment across manipulation conditions. (a) Mean target selection rate (i.e., proportion of trials in which participants attributed the synthesized image to the target identity B) aggregated by the number of manipulated features. The dashed line marks the 50% ambiguity boundary (chance level in a forced-choice task). (b) Heatmap of target selection rates for humans and AI similarity models across manipulation conditions; values indicate the proportion of stimuli classified as the target B for each row/condition.

($\beta = 1.41$); face shape showed no significant effect (Table 1). When only eyes were manipulated, 39.9% of stimuli were perceived as the target, increasing sharply with feature combinations (eyes+nose: 70.5%, eyes+nose+mouth: 87.7%). In contrast, manipulations excluding eyes yielded target selection rates below 5%. These findings aligned with participants’ subjective reports. In a post-survey open-ended question asking which features informed their judgments, 79.6% of participants mentioned eyes—substantially exceeding the next most frequent responses: overall impression (34.9%) and face shape (24.2%). This pattern held regardless of familiarity. Despite prior research indicating differential reliance on internal versus external features by familiarity, we observed no significant difference (familiar $\beta = 4.21$, unfamiliar $\beta = 4.78$), suggesting that identity perception in deepfake contexts relies on the eye region irrespective of familiarity.

3.2 The Identity Threshold: When Two Features Change

To further address RQ1, we complemented the feature-level GLMM results (Section 3.1; which cues matter) with an analysis that asks a different question: when does identity perception shift as more features are manipulated, and do combined changes follow additivity? We aggregated target selection rates by the number of manipulated features (Figure 2a). Target selection increased from 10.1% (one feature) to 51.3% (two features), surpassing the 50% ambiguity boundary, and reached 87.7% with three features. This shift was not explained by linear accumulation: eye-inclusive pairs showed super-additive effects relative to additive predictions (eyes+nose: 70.5% vs. 42.6%; eyes+mouth: +22.2pp), whereas nose+mouth remained below the ambiguity boundary (18.6%) with a smaller super-additive effect (+13.0pp). Overall, eyes show both a strong main effect and interaction effects that amplify identity perception shifts when combined with other features.

3.3 Human-AI Misalignment in Identity Judgment

Figure 2b visualizes human-AI alignment by plotting target selection rates for humans and AI models under each manipulation condition. Marked differences emerged between human and AI

judgments. For single-feature manipulations, humans exhibited strong responses only to eyes (0.40), showing minimal response to other features (0.01–0.04). AI models, in contrast, maintained similar selection rates regardless of which feature was manipulated. Pronounced misalignment occurred in eye-inclusive combined manipulations (eyes+nose, eyes+mouth). Humans clearly perceived these as the target (0.71, 0.65), whereas face recognition models showed lower rates: ArcFace (0.35, 0.31), FaceNet (0.63, 0.51), and VGG-Face (0.56, 0.47). Although OpenFace and CLIP exhibited human-like rates in these conditions, their uniformly high selection across all conditions precludes alignment with human judgment patterns. Importantly, these conditions correspond to the two-feature identity threshold identified earlier, representing the region of most pronounced human-AI divergence. This suggests that current AI-based detection systems risk failing to flag synthesized content that humans perceive as the victim.

4 Discussion

Summary and Theoretical Implications. This study explored the conditions under which synthesized faces are perceived as specific individuals in deepfake contexts, examining alignment between human identity perception and AI similarity models. For RQ1, we found that eyes serve as the dominant cue for identity perception, with the identity threshold reached when two features, including eyes, are manipulated simultaneously. For RQ2, human-AI misalignment was most pronounced at this threshold region. Crucially, our findings suggest that identity violations are not defined by how much a face is manipulated, but by when perceptual attribution crosses a threshold.

These findings extend cognitive psychology’s eye-centric processing theory to deepfake contexts while revealing an intriguing divergence. Prior research indicates that familiar and unfamiliar faces follow distinct cognitive pathways; however, in our study, eyes showed consistent influence regardless of familiarity. This familiarity-independent pattern suggests that identity judgment in deepfake contexts may constitute a distinct task from general face recognition.

Furthermore, the super-additive effects observed in eye-inclusive combined manipulations carry practical implications in the deepfake context. Even when deepfake creators manipulate only eyes and one or two additional features, humans may perceive the synthesized face as the victim, with identity shift occurring in a nonlinear rather than additive manner. Such patterns suggest that deepfake detection systems must account for feature interactions, and that limiting the scope of manipulation does not prevent harm.

Practical Implications for Detection and Regulation. Our findings show direct implications for deepfake detection and regulatory design. Face recognition models widely used for measuring identity similarity in deepfake research tended to miss synthesized faces that humans perceived as the target at the two-feature threshold. This reveals a critical gap: humans perceive the victim, but AI does not—suggesting that AI-based detection systems risk overlooking deepfake content with the potential to cause real harm.

Detection systems may require greater weighting of eye-region changes or calibration that reflects human perception patterns.

From a regulatory perspective, our findings provide empirical grounding for the legal gap of “what level of similarity constitutes identity infringement.” Identifying two-feature manipulation as the identity confusion threshold may serve as a reference point for defining harm criteria in future deepfake legislation. Ultimately, since deepfake harm arises from human perception rather than technical manipulation alone, a victim-centered approach is warranted in both detection and regulation. Together, these implications suggest that effective responses to deepfake harm must account for how identity is perceived, rather than relying solely on technical similarity or binary authenticity judgments.

Limitations and Future Work. This study has several limitations. First, stimuli were manually created using GIMP, which may not fully capture artifacts or characteristics of actual deepfake generation algorithms. Second, we used only static images, failing to capture the temporal dynamics of video-based deepfakes. Future work should validate these findings using GAN- or diffusion-based deepfake syntheses and video stimuli. Despite these limitations, this study provides a foundation for future research and practice by offering the first systematic exploration of identity perception thresholds in deepfakes and human-AI misalignment patterns.

5 Conclusion

This study explored threshold conditions for identity perception in deepfakes and human-AI alignment. Eyes serve as the dominant identity cue, with the identity threshold reached when two features, including eyes, are manipulated. Human-AI misalignment peaks at this threshold, suggesting current detection systems risk missing content humans perceive as harmful, particularly in eye-inclusive two-feature manipulations where identity attribution shifts most sharply. These findings call for detection systems that prioritize changes in the eye region and feature interactions, alongside victim-centered regulatory approaches. As the first systematic exploration of identity thresholds and human-AI misalignment in deepfakes, this work provides a foundation for future research and practice, informing how identity-related harm may be operationalized around perceptual thresholds rather than manipulation extent alone.

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